Evaluating Multi-Document Retrieval-Augmented Generation for Student Financial Aid Guidance: Authority Confusion, Temporal Confusion, and the Limits of Faithfulness

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Abstract—**Large language models deployed for student financial aid guidance risk producing fluent but incorrect answers, with direct consequences for first-generation and low-income students. We evaluate whether retrieval-augmented generation (RAG) reduces this risk on a 55-question benchmark across five question categories, drawn from a 1,641-chunk corpus of federal aid handbooks (2024–25 and 2025–26), consumer guidance, and three institutional policy pages. We compare a no-retrieval baseline against standard and authority-aware RAG across 165 runs, scoring for Faithfulness, Answer Correctness, and Source Authority Accuracy. Three findings emerge. First, RAG improves mean Answer Correctness over baseline (0.545 vs. 0.255) but the gain vanishes on contradictory-source questions. Second, correctness degrades with required source count, consistent with a multi-hop integration cost. Third, Faithfulness exceeds Answer Correctness by roughly 0.40 across all RAG runs: a systematic gap illustrated by one question where a fully grounded answer is factually wrong because re-ranking promoted a methodology chunk over the fact-bearing chunk in the same document. Authority-aware re-ranking resolves temporal conflicts but suppresses institutional sources on the question type it was not designed for. We contribute an annotated benchmark, a three-condition pipeline, and empirical evidence that retrieval correctness and answer correctness are not interchangeable in high-stakes informational domains.**

I. Introduction

On the first day of this project, a hallucination audit of ten representative student financial aid questions was run against a bare large language model with no retrieval support. The model answered six of ten questions correctly, partially answered three, and never fabricated a number outright. That single result was deceptive: every correct answer came from a federal rule that had not changed in the model's training window, and the model had no way to know whether it was reciting current policy or a rule that had since been superseded. The audit could not distinguish a model that knew the answer from a model that had memorized last year's answer.

This is the central problem facing any AI system deployed for financial aid guidance. The Free Application for Federal Student Aid (FAFSA) Simplification Act rewrote the formula for federal aid eligibility for the 2024–25 award year, replacing the Expected Family Contribution with the Student Aid Index and changing how family size and multiple-student households are treated. Individual universities layer institutional policy on top of the federal floor, sometimes more strictly, as when a university requires an 80% credit completion rate against a federal minimum of 67%. A student asking an AI assistant whether they meet the requirements for continued aid is asking a question whose correct answer depends on which year's rule applies and which authority, federal or institutional, governs the specific aid program in question. A language model with no access to current, source-attributed documents cannot reliably answer either question.

The gap this work addresses is not whether retrieval-augmented generation improves factual grounding in general; that has been established across general-knowledge benchmarks. The gap is whether RAG, and specifically an authority-aware variant designed to resolve conflicting and outdated sources, performs reliably in a domain where (1) multiple documents must sometimes be integrated to answer a single question, (2) institutional and federal sources legitimately disagree, and (3) the correct answer changes from one year to the next within the same document set. No published evaluation addresses all three conditions simultaneously in the financial aid domain, and no prior multi-hop RAG benchmark is built around legitimately conflicting rather than merely distractor sources.

This paper makes four contributions. First, a 55-question benchmark stratified into five categories, single-source, two-source, three-or-more-source, contradictory-source, and temporal, each annotated with verbatim ground truth, required source documents, and, for contradictory and temporal questions, the specific conflicting or superseded text. Second, a working three-condition RAG pipeline (no-retrieval baseline, standard RAG, authority-aware RAG) evaluated against a real, modified corpus of federal handbooks, consumer guidance, and institutional policy pages. Third, an empirical comparison of three evaluation metrics, automated Faithfulness, human-scored Answer Correctness, and rule-based Source Authority Accuracy, that demonstrates these metrics measure different things and can move in opposite directions. Fourth, a specific, reproducible demonstration of authority-aware re-ranking actively harming performance on the question category it was not designed for, informing a concrete design recommendation against uniform priority re-ranking.

The remainder of this paper is organized as follows. Section II reviews related work in multi-hop RAG benchmarking, RAG in educational applications, and financial aid AI systems, and identifies the gap each leaves open. Section III describes the system architecture, corpus, and question taxonomy. Section IV defines the three evaluation metrics and the statistical analysis plan. Section V reports results organized by research question. Section VI discusses the findings through the lens of three failure modes and offers four design recommendations. Section VII concludes with answers to each research question, contributions, and future directions.

II. Related Work

A. Multi-Hop RAG Benchmarks

The dominant multi-hop question-answering benchmarks, HotpotQA [1], 2WikiMultiHopQA [2], and MuSiQue [3], evaluate whether a RAG system can combine evidence from multiple Wikipedia passages to answer a single question. HotpotQA questions are drawn from general-knowledge Wikipedia content and have been criticized for permitting answer shortcuts that bypass genuine multi-hop reasoning [4]. 2WikiMultiHopQA addresses this by combining unstructured text with structured Wikidata relations, and MuSiQue enforces connected multi-hop chains by composing questions bottom-up from single-hop components [3]. More recent work, including MultiHop-RAG [5] and retrieval-focused agentic frameworks evaluated across these same three benchmarks [6], has extended the evaluation methodology but not the domain: every benchmark in this line of work draws its corpus from general encyclopedic knowledge or, in MultiHop-RAG's case, a news-article corpus, where evidence is fundamentally non-conflicting; multiple sources support a single converging answer rather than disagreeing with one another. None of these benchmarks include questions where two correctly-retrieved sources legitimately give different answers, and none model a corpus where the same governing document changes from one time period to the next while both versions remain in the index. **Gap: existing multi-hop RAG benchmarks test evidence integration over non-conflicting general-knowledge or news corpora, leaving authority conflict and temporal versioning within a single domain corpus unaddressed.**

B. RAG in Education

A growing body of work applies RAG to educational support, including course-specific tutoring assistants grounded in lecture material [12], programming-course tutors [13], K-12 content support [14], and automated essay feedback systems that report high agreement with human graders [15]. Survey work on RAG for education [16] catalogs these applications and consistently identifies hallucination and static internal knowledge as the central motivations for adding retrieval, echoing concerns raised in programming-education contexts where LLMs are shown to mislead learners without external grounding [17]. Personalized intelligent tutoring systems built on RAG and prompt engineering [18] and chatbot tutors deployed in higher-education courses [19] report improved relevance and engagement metrics. None of this literature, however, evaluates a setting where the underlying institutional rules a student is asking about can themselves be in conflict (a university policy stricter than a federal minimum) or can change from one academic term to the next while older material remains retrievable. Educational RAG evaluation to date measures content relevance and engagement, not authority resolution or temporal correctness. **Gap: RAG-in-education research has not evaluated systems against corpora containing legitimate authority conflicts or version-dependent correct answers, both of which are common in regulatory and policy-governed student-facing domains.**

C. Financial Aid AI Systems

Deployed financial aid AI takes the form of commercial FAQ chatbots and rule-based workflow automation. Vendors report chatbots that answer FAFSA status questions, monitor application deadlines in real time [20], and route routine eligibility checks via predefined rules [21], with some institutions reporting expanded after-hours support coverage from virtual assistants [22]. Other commercial products are explicitly described as standardized-guidance or FAQ-search systems trained on an institution's own website content [23] rather than as systems that reason over multiple authoritative documents. None of this deployed technology has been evaluated as a multi-document RAG system against a benchmark with annotated ground truth, and none of the public descriptions of these systems address how they would resolve a conflict between a federal rule and an institution's own stricter policy, or between this year's rule and last year's. **Gap: existing financial aid AI is rule-based or FAQ-retrieval in design and has not been evaluated as a multi-document RAG system against an annotated benchmark covering authority conflict or temporal versioning.**

III. System and Dataset

A. Pipeline Architecture

The system is implemented as six independent modules, each unit-tested in isolation before being wired into the full experimental loop.

M1 (Indexer) loads all corpus text files from five authority-labeled folders (federal-primary, federal-outdated, federal-consumer, professional, institutional), chunks them, embeds each chunk with `all-MiniLM-L6-v2`, and confirms the resulting ChromaDB collection is complete and queryable against a test query before any experimental run begins.

M2 (Retriever) accepts a query string and returns the top-$k = 5$ chunks by cosine similarity, each annotated with chunk ID, document name, source authority tag, award year, and similarity distance. When fewer than 5 chunks meet the similarity threshold, M2 returns all available chunks; in practice this never occurred for the 55 experimental queries, though the small institutional tier (23 chunks) means that queries specifically targeting institutional content may retrieve fewer than 5 institutional chunks even when the institutional source is present in the index, which contributes to the C-01 and C-02 retrieval failures discussed in Section V-C.

M3 (Re-ranker), used only in the RAG-AuthAware condition, accepts the top-5 chunks from M2 and a target award year, then re-sorts them according to a fixed priority order: federal-primary chunks matching the target year receive priority 1; federal-consumer receives priority 2; professional receives priority 3; institutional receives priority 4; and federal-primary chunks *not* matching the target year are demoted to priority 5, the lowest rank. Formally, for chunk c with authority $a(c)$ and award year $y(c)$, the re-ranking priority $\pi(c)$ is

$$\pi(c) = \begin{cases} 1 & a(c) = \text{federal-primary},\ y(c) = y_{\text{target}} \\ 5 & a(c) = \text{federal-primary},\ y(c) \neq y_{\text{target}} \\ P(a(c)) & \text{otherwise} \end{cases} \tag{1}$$

where $P(\cdot)$ maps each remaining authority to its fixed priority (2 for federal-consumer, 3 for professional, 4 for institutional). Chunks are sorted by $(\pi(c), d(c))$, breaking ties by the original cosine distance $d(c)$.

M4 (Prompt builder) formats the (re-ranked or unranked) chunks as numbered context blocks, each labeled with its source document, authority, and award year, and prepends the instruction: *"Answer using ONLY the information in the context below."* For the Baseline-LLM condition, no context is supplied and the question is passed directly.

M5 (LLM caller) submits the assembled prompt to `gpt-4o-mini` at temperature 0, recording the model name, the full version string returned by the API, and prompt/completion/total token counts for every call.

M6 (Logger) appends one row per run to `raw_log.csv`, with 16 fields: run ID, question ID, category, condition, query text, retrieved chunk IDs, retrieved authority tags, retrieved award years, retrieved document names, the generated answer, the Faithfulness score, prompt/completion/total token counts, model version, and timestamp.

B. Corpus

The corpus comprises 1,641 chunks (400 tokens, 50-token overlap, `o200k_base` tokenizer) drawn from five authority tiers, summarized in Table I.

TABLE I
CORPUS COMPOSITION (TABLE I)

Source Authority	Files	Chunks	Avg. Tokens/Chunk
Federal primary (2025–26)	30	785	343
Federal outdated (2024–25)	30	786	341
Federal consumer	6	41	339
Professional (NASFAA)	1	6	379
Institutional (3 universities)	4	23	316
Total	**71**	**1,641**	**345**

Federal-primary documents are the 2025–26 Federal Student Aid Handbook; federal-outdated documents are the equivalent 2024–25 Handbook, retained in the index to create genuine temporal conflicts rather than simulating them. Federal-consumer documents are studentaid.gov guidance pages. Institutional documents are Satisfactory Academic Progress (SAP) policy pages from three public universities, selected specifically because each contains at least one rule stricter than the corresponding federal minimum (e.g., an 80% credit-completion requirement against a 67% federal floor, or a 2.5 GPA requirement against a 2.0 federal floor). A fourth candidate professional-tier document (a national aid statistics profile) was removed after a Week-2 retrieval audit found it was dominating retrieval results for broad queries without containing question-specific eligibility content; this removal is the corpus's only post-hoc modification and is documented for reproducibility.

C. Question Taxonomy

All 55 questions are written in first-generation-student lay language and assigned to exactly one of five categories, summarized with one example each below.

S (Single-source, $n = 15$**)**: the complete answer exists in one chunk from one document. *Example: "What is the maximum Pell Grant I can get for the 2025–26 school year?"* Ground truth ($7,395) is fully contained in the 2025–26 Federal Student Aid Handbook, Volume 7, Chapter 2.

M2 (Two-source, $n = 15$**)**: neither source alone is sufficient; both are required. *Example: "If I take out a subsidized loan, do I have to pay interest while I'm still in school?"* Requires combining the consumer-facing loan-type page (which states the government covers interest while enrolled) with the Handbook's interest-accrual provisions (which specify the half-time enrollment threshold this depends on).

M3 (Three-or-more-source, $n = 10$**)**: at least three sources are jointly necessary and no two-source subset suffices. *Example: "Can I get a Pell Grant, a Work-Study job, and a loan all at the same time, and is there a limit to how much I can get total?"* Requires the Handbook's packaging chapter, the consumer Pell Grant overview, and the consumer Work-Study overview; omitting any one leaves the question only partially answered.

C (Contradictory-source, $n = 5$**)**: at least one institutional document conflicts with the federal answer. *Example: "Does the government require me to pass 80% of my classes to keep my financial aid?"* Ground truth is *no*; the federal Satisfactory Academic Progress standard is 67% completion, but the University of Washington's institutional policy states verbatim, "Complete a minimum of 80% of the cumulative credits you attempted," a stricter institutional rule that a retrieval system might surface as if it were the federal answer.

T (Temporal, $n = 10$**)**: the correct answer changed between the 2024–25 and 2025–26 award years. *Example: "How does the government figure out how much financial aid I need?"* The 2024–25 correct answer used the Expected Family Contribution (EFC) formula; the 2025–26 correct answer uses the Student Aid Index (SAI), which, unlike the EFC, can be negative, under the FAFSA Simplification Act.

D. Experimental Conditions

Three conditions are run for every question, for a total of $55 \times 3 = 165$ evaluations.

Baseline-LLM: the question is submitted directly to `gpt-4o-mini` with no retrieved context, isolating what the model's training data alone supports.

RAG-Standard: the top-5 chunks returned by M2 (cosine similarity over the full mixed corpus, with no authority awareness) are passed to M4 and M5 unmodified.

RAG-AuthAware: the top-5 chunks from M2 are passed through the M3 re-ranking algorithm described in Eq. 1 before being passed to M4 and M5, promoting current-year federal sources and demoting outdated federal sources, without regard to question category.

IV. Experimental Design and Metrics

A. Metric Definitions

Faithfulness (F) is computed by decomposing the generated answer into atomic factual claims via an LLM call, then checking each claim against the retrieved context chunks via a second LLM call, and reporting the proportion of claims supported:

$$F = \frac{|\{\text{claims supported by context}\}|}{|\{\text{all claims in answer}\}|}, \quad F \in [0, 1] \qquad (2)$$

$F = 0$ is assigned directly, without an LLM call, whenever no context was supplied (the Baseline-LLM condition) or the answer is empty.

Answer Correctness (A) is scored by human annotation against the verbatim ground truth in the question bank, on a three-point scale: $A = 0$ (wrong), $A = 0.5$ (partially correct, right direction but missing a required detail or citing an incorrect threshold), or $A = 1$ (correct). For C-category questions, correctness is defined relative to the federal (priority-1) answer; for T-category questions, correctness is defined relative to the 2025–26 rule.

Source Authority Accuracy (S) is a rule-based binary score: $S = 1$ if and only if the question's annotated expected source authority appears among the retrieved authority tags for that run; $S = 0$ otherwise. Baseline-LLM is assigned $S = 1$ by convention, as no retrieval occurs and the metric is not applicable. Critically, S does not check the rank position of the expected authority within the top-5, only its presence; this limitation is returned to in Section VI.

B. Statistical Analysis Plan

A two-way ANOVA is run independently for each metric using the model metric $\sim$ Condition + Category + Condition $\times$ Category (Type II sums of squares, `statsmodels`), with $df_{\text{Condition}} = 2$, $df_{\text{Category}} = 4$, $df_{\text{Condition}\times\text{Category}} = 8$, and $df_{\text{Residual}} = 150$. This model tests the main effect of condition (RQ1: does RAG outperform baseline?) and the Condition$\times$Category interaction (RQ3: does the RAG benefit differ across question types?). A supplementary one-way ANOVA on category alone ($df = 4, 160$, pooling conditions) is retained in the statistical supplement for descriptive comparison but is not the primary inferential test. A linear regression of Answer Correctness on source count (S$\rightarrow$1, M2$\rightarrow$2, M3$\rightarrow$3) is fit separately within each of the three conditions, testing the hypothesis that correctness declines monotonically as more sources must be integrated. Bonferroni-corrected paired t-tests compare RAG-Standard against RAG-AuthAware within each of the five categories, with the family-wise significance threshold set at $\alpha_{\text{corr}} = 0.05/3 = 0.0167$ to control for the three primary comparisons of theoretical interest (S, C, and T categories specifically motivate the re-ranker; M2 and M3 are included for completeness).

C. Inter-Rater Protocol

A stratified 25% sample of all human-scored Answer Correctness annotations ($n = 42$, approximately 8–9 per category, drawn with a fixed random seed for reproducibility) was independently re-scored by the supervising researcher using the same three-point rubric. Cohen's κ was computed via `sklearn.metrics.cohen_kappa_score` treating the three score levels as categorical labels. The pre-registered threshold for acceptable annotation reliability was $\kappa \geq 0.70$; at or above this threshold, the full annotation set is treated as reliable and any remaining disagreements on the sample are resolved by discussion with reference to the original ground-truth text, with each resolution documented. Below threshold, the annotation protocol would be considered ambiguous for at least one category, requiring protocol revision and re-scoring of the affected category before the full results could be reported. Inter-rater agreement on the stratified 42-answer sample was Inter-rater agreement on the stratified 42-answer sample was $\kappa = 0.887$, exceeding the pre-registered threshold of 0.70 and indicating almost perfect agreement. Three disagreements were resolved by discussion with reference to the ground-truth text and documented.

V. Results

A. RQ1: Does RAG Improve Correctness and Grounding Over Baseline?

Figure 2 reports mean Answer Correctness by category and condition. Baseline-LLM scored a mean $A = 0.255$ across all 165 runs, against $A = 0.545$ for RAG-Standard and $A = 0.509$ for RAG-AuthAware, a gain that is largest on S ($0.267 \rightarrow 0.700$) and M2 ($0.367 \rightarrow 0.600$) and absent on C, where Baseline-LLM and RAG-Standard tie at $A = 0.200$ and RAG-AuthAware falls to $A = 0.100$. A two-way ANOVA (Condition $\times$ Category) confirms a significant main effect of Condition on Answer Correctness, $F(2, 150) = 8.87$, $p < 0.001$, establishing that RAG improves correctness over baseline. The main effect of Category is also significant, $F(4, 150) = 3.32$, $p = 0.012$, confirming that question type systematically affects correctness. Critically, the Condition$\times$Category interaction is not significant,

$F(8,150) = 0.51$, $p = 0.847$: the RAG improvement over baseline is statistically consistent across all five categories, meaning the C-category drop visible in Figure 2 is a descriptive pattern rather than a statistically reliable differential effect at this sample size. The same two-way ANOVA on Faithfulness yields a dominant main effect of Condition, $F(2,150) = 875.47$, $p < 0.001$, a significant Category effect, $F(4,150) = 3.78$, $p = 0.006$, and a non-significant interaction, $F(8,150) = 1.08$, $p = 0.382$. For Source Authority Accuracy, the Condition main effect is significant, $F(2,150) = 4.00$, $p = 0.020$, Category is marginal, $F(4,150) = 2.44$, $p = 0.050$, and the interaction is not significant, $F(8,150) = 0.61$, $p = 0.769$. Pairwise, RAG-Standard's correctness advantage over RAG-AuthAware on C-category questions (0.200 vs. 0.100, a 50% relative drop) does not reach Bonferroni-corrected significance, $t(4) = 1.00$, $p = 0.374$, against $\alpha_{\text{corr}} = 0.0167$; this null result should be interpreted as inconclusive rather than as evidence of no effect, given the very low statistical power at $n = 5$ per condition within this category.

B. RQ2: Does Correctness Degrade With Source Count?

Figure 3 plots mean Answer Correctness against source count (1, 2, or 3) for all three conditions. RAG-Standard shows the steepest decline, from $A = 0.700$ at one source to $A = 0.400$ at three sources, a regression slope of -0.146 ($R^2 = 0.075$, $p = 0.087$); RAG-AuthAware shows a comparable decline from $A = 0.633$ to $A = 0.400$ (slope -0.113, $R^2 = 0.042$, $p = 0.205$); Baseline-LLM shows the shallowest and non-monotonic trend (slope -0.046, $R^2 = 0.008$, $p = 0.583$), rising slightly from one to two sources before falling at three. None of the three regressions reaches significance at $\alpha = 0.05$, though RAG-Standard's slope is the closest to threshold and is also the largest in magnitude of the three conditions, consistent with retrieval-dependent conditions having more to lose as integration demands increase. The paired t-test comparing RAG-Standard and RAG-AuthAware on M3 questions yields $t(9) = 0.000$, $p = 1.000$, confirmed by per-question inspection: every one of the ten M3 questions received identical Answer Correctness scores under both conditions, with zero exceptions. This is not a numerical artifact — it is a substantive finding. Authority-aware re-ranking alters Faithfulness on three M3 questions (M3-01: 0.875 vs. 1.000; M3-02: 0.800 vs. 1.000; M3-08: 1.000 vs. 0.800) but does not change whether any M3 answer is ultimately correct. The same pattern holds for T-category questions ($t(9) = 0.000$, $p = 1.000$, all ten questions identical in Answer Correctness despite Faithfulness differences on T-01, T-03, and T-05), suggesting that for multi-source integration and temporal questions, re-ranking modifies context grounding but the underlying answer judgment is robust to the specific chunk ordering within the top-5.

C. RQ3: Do Conflicting and Outdated Sources Cause Errors, and Does Re-Ranking Help?

Figure 4 isolates Source Authority Accuracy for the C and T categories, the two categories where authority-aware re-ranking was hypothesized to matter most. On T-category questions, RAG-Standard and RAG-AuthAware are tied at $S = 0.800$; on C-category questions, both conditions are tied at $S = 0.600$. Despite this tie on S, Answer Correctness on C-category diverges: RAG-Standard scores $A = 0.200$ against RAG-AuthAware's $A = 0.100$, a pattern investigated directly by retrieving the underlying chunk order for all five C-category questions. In two of five cases (C-03 and C-05), the institutional source containing the correct contradicting evidence was retrieved at rank 1 under RAG-Standard but pushed to rank 2 or rank 3 under RAG-AuthAware's fixed federal-priority promotion; in one case (C-04, where the federal source is in fact the correct answer), re-ranking correctly promoted the federal chunk from rank 4 to rank 1; in the remaining two cases (C-01, C-02) the institutional source never appeared in the top-5 under either condition, a retrieval failure unrelated to re-ranking. S, which only checks presence in the top-5 and not rank position, is accordingly unable to distinguish a re-ranking-induced position regression from a genuine retrieval success.

D. Faithfulness ≠ Correctness

Figure 1 reports mean Faithfulness by category and condition. Across all 165 runs, RAG-Standard's mean Faithfulness ($F = 0.929$) and RAG-AuthAware's ($F = 0.938$) both substantially exceed their respective mean Answer Correctness scores (0.545 and 0.509), a gap of roughly 0.40 in both conditions. The single clearest case is question S-01 ("What is the maximum Pell Grant I can get for the 2025–26 school year?") under RAG-AuthAware: the system scores $F = 1.00$ (every claim in the answer is grounded in the retrieved context) and $A = 0$ (the answer is wrong). Inspection of the retrieved chunks shows the re-ranker promoted a methodology chunk from the correct 2025–26 Handbook volume, one that explains how the maximum award is determined without stating the $7,395 figure itself, ahead of a consumer-facing chunk that does contain the dollar figure and was in fact ranked first under RAG-Standard (where the same question scored $A = 1$). The error originates in chunk-boundary placement at the corpus-construction stage, amplified by authority promotion, and is not detectable by the Faithfulness metric, which only checks whether claims are grounded in whatever context was supplied, not whether the supplied context was sufficient.

E. Error Taxonomy

All 74 answers scored $A = 0$ across all three conditions were classified into one of four failure types: retrieval (the correct document was absent from the retrieved set), generation (the correct document was retrieved but the model failed to use it correctly), authority (a lower-priority source's answer was reported as the correct answer), or temporal (a superseded rule was reported as current). The category totals are: retrieval = 24, generation = 39, authority = 2, temporal = 9, summing to 74. Of the 37 zero-scored answers from the two RAG conditions specifically, retrieval failures (24) and generation failures (9) are the largest categories, with 2 authority-confusion cases (both on S-06) and 2 temporal cases. The remaining 37 zero-scored answers belong to Baseline-LLM, where all non-temporal failures reflect training-data knowledge gaps rather than retrieval pipeline errors. Temporal-pattern failures account for 9 of 74 zero-scored answers across all conditions and are concentrated entirely in the T category, consistent with the category-level finding in RQ3.

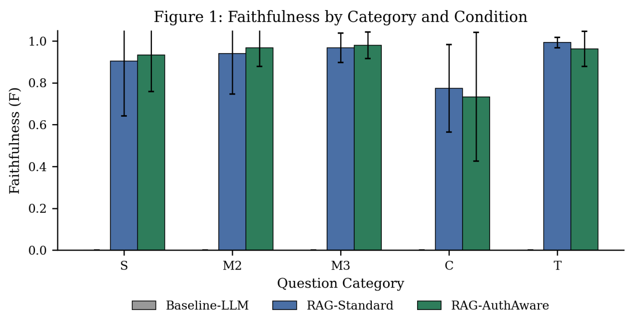

Fig. 1. Mean Faithfulness by question category (S, M2, M3, C, T) and experimental condition. Error bars show one standard deviation. Baseline-LLM is fixed at $F = 0$ in every category since no context is supplied. Both RAG conditions exceed $F = 0.9$ in four of five categories, including C and T, despite the much lower Answer Correctness reported for those same categories in Fig. 2, illustrating the Faithfulness $\neq$ Correctness gap discussed in Section V-D.

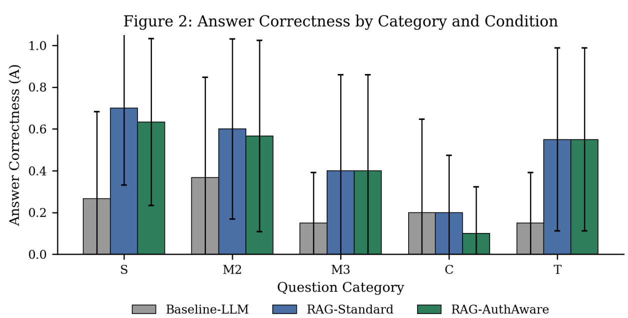

Fig. 2. Mean Answer Correctness by question category (S, M2, M3, C, T) and experimental condition (Baseline-LLM, RAG-Standard, RAG-AuthAware). Error bars show one standard deviation. RAG-AuthAware underperforms RAG-Standard specifically on the C category, the category the re-ranker was not designed to handle.

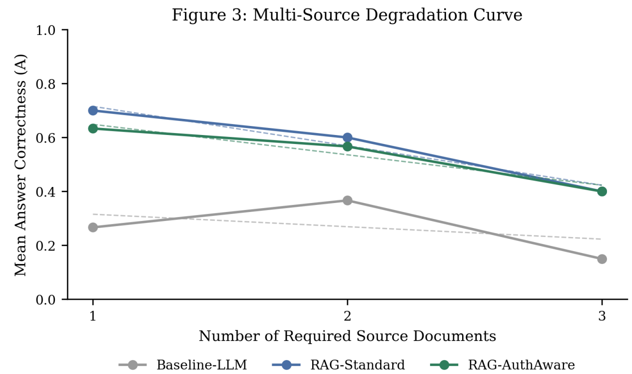

Fig. 3. Mean Answer Correctness as a function of the number of source documents required to fully answer the question (1 = single-source, 2 = two-source, 3 = three-or-more-source), with linear regression lines overlaid for each condition. Both RAG conditions decline more steeply than Baseline-LLM, though no slope reaches significance at $\alpha = 0.05$.

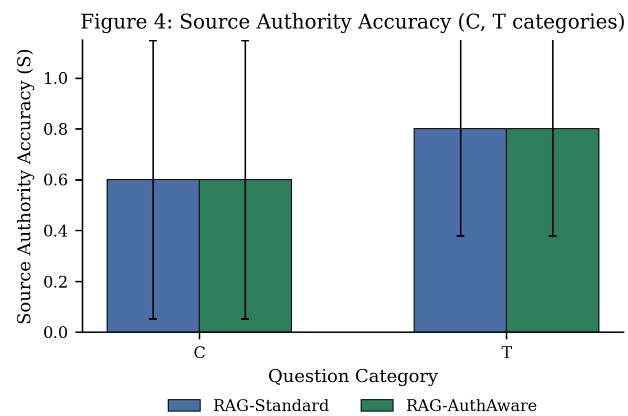

Fig. 4. Source Authority Accuracy for the C (contradictory-source) and T (temporal) categories only, comparing RAG-Standard against RAG-AuthAware. Error bars show one standard deviation. The two conditions are statistically tied on this metric in both categories, despite diverging on Answer Correctness within the C category (see Fig. 2).

VI. Discussion

A. Findings Through the Lens of Three Failure Modes

Multi-source integration cost. The RQ2 regression results, while not individually significant, are directionally consistent across both RAG conditions and point toward a genuine integration cost: each additional required source appears to subtract roughly 13–15 percentage points of correctness. This is plausible given the pipeline design, since a single retrieval pass of $k = 5$ chunks must increasingly subdivide its attention across more required documents as the question's source count grows, with no mechanism to verify all required sources were actually retrieved before generation begins.

Authority confusion. The S-06 case (Section V-E) and the C-category position-shift findings (Section V-C) together describe the same underlying failure mode from two directions. In S-06, an institutional chunk states a stricter, university-specific threshold without the corpus context to disambiguate it from the federal standard, and the LLM treats it as authoritative. In C-03 and C-05, the institutional source is the one the question requires, and the re-ranker's fixed priority order actively suppresses it. Both failures stem from the same root cause: the system has no mechanism for the model or the re-ranker to recognize *when* institutional authority should dominate versus when federal authority should dominate, because that determination depends on the question's intent, not on a fixed document property.

Temporal confusion. All 9 temporal-pattern error-taxonomy cases occur in the T category, where outdated 2024–25 material remains in the index alongside the current 2025–26 version. RAG-AuthAware's demotion of outdated federal-primary chunks to rank 4–5 (Eq. 1) is the correct general-purpose fix for this failure mode, and the tied $S = 0.800$ for both RAG conditions on T-category understates this benefit, since S does not capture rank position; a systematic chunk-order audit of all 10 T-category questions analogous to the C-category audit in Section V-C would be required to quantify this benefit precisely; this analysis is identified as a future direction in Section VII.

B. Design Recommendations

(1) *Category-aware re-ranking.* The single fixed priority order used here helps T-category questions and actively harms a subset of C-category questions. A production system should condition the re-ranking priority on a lightweight question-type classifier, suppressing the federal-priority promotion specifically when the question is detected as contradiction-seeking.

(2) *Chunk-level answer verification before generation.* The S-01 failure (Section V-D) occurred because the correct document was retrieved but the correct *chunk* within that document was not. A verification step that checks whether the retrieved context actually contains an answerable fact, not merely a topically related passage, before passing it to the generator would catch this class of error before it reaches the user.

(3) *Multi-source completeness checking.* Given the RQ2 degradation pattern, a system answering M3-category-style questions should explicitly verify that all annotated required source types have been retrieved before generation, falling back to an "insufficient information" response rather than a partial answer when one is missing.

(4) *Separating retrieval success from rank quality in evaluation.* The Source Authority Accuracy metric used here is a presence check, not a rank check, and as a result it failed to distinguish a genuine re-ranking improvement from a genuine re-ranking regression in the C category. Future evaluation of authority-aware retrieval should report a rank-weighted variant of this metric, such as reciprocal rank of the expected authority, to make this distinction visible without requiring manual chunk-order inspection.

C. Limitations

Six limitations qualify these findings. First, all three conditions use a single LLM (`gpt-4o-mini`); results may not generalize to larger or differently-trained models, which could be more or less susceptible to the chunk-selection and authority-confusion failures observed here. Second, the institutional corpus draws from three universities; the specific conflicts identified (80% completion, 2.5 GPA) are real but may not represent the full range of institutional policy variation nationally. Third, the corpus and benchmark are English-only, and the failure modes identified may interact differently with multilingual retrieval or non-English source documents. Fourth, Answer Correctness was scored by human annotation with only a 25% independently-verified sample; while inter-rater agreement was assessed (Section IV-C), full-dataset annotation subjectivity, particularly for partially-correct (0.5) judgments, remains a source of measurement noise. Fifth, the institutional and outdated-federal portions of the corpus, while drawn from real published policy pages, were assembled and curated specifically for this study rather than reflecting an organically occurring production corpus, and the corpus's small institutional tier (23 chunks) may not reflect retrieval competition dynamics at larger scale. Sixth, no user study was conducted; whether real students would notice, trust, or act on the specific failure modes documented here (a faithful-but-wrong answer, an institutional rule reported as federal) is an open empirical question this work does not address.

VII. Conclusion

RQ1. RAG improves mean Answer Correctness over a no-retrieval baseline (0.545 and 0.509 vs. 0.255), but this improvement is category-dependent and disappears entirely on contradictory-source questions, where authority-aware re-ranking in fact underperforms both alternatives.

RQ2. Answer Correctness declines as the number of required source documents increases from one to three across all three conditions, with the steepest decline in RAG-Standard, though no individual regression slope reaches conventional statistical significance at this sample size.

RQ3. Authority-aware re-ranking measurably helps resolve temporal conflicts but is tied with standard RAG on Source Authority Accuracy in the contradictory-source category specifically because the presence-only metric cannot detect that re-ranking demonstrably pushed the correct institutional source down in retrieval rank for two of five contradictory-source questions.

This work contributes: (1) a 55-question, five-category benchmark for financial aid RAG evaluation with verbatim ground truth and documented source conflicts; (2) a working three-condition RAG pipeline with a precisely specified authority re-ranking algorithm (Eq. 1); (3) an empirical demonstration that Faithfulness and Answer Correctness are not interchangeable and can diverge by a full point on the same question; and (4) a concrete finding that uniform authority-based re-ranking is a double-edged design choice, beneficial for the temporal-conflict case it was designed for and actively harmful for a subset of the contradictory-source case it was not.

Four directions follow directly from the limitations above: extending the benchmark to multiple LLMs to test whether the chunk-selection and authority-confusion failures are model-specific or architectural; building and evaluating the category-aware re-ranker proposed in Section VI-B; extending Source Authority Accuracy to a rank-sensitive variant capable of detecting re-ranking regressions without manual inspection; and conducting a user study to determine whether real students can detect a faithful-but-incorrect answer of the kind documented in Section V-D.

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